Knowing When You Are Wrong: Validated Recalibration, Pose-Risk Gating and World Models for Reliable Simulated Industrial Picking

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Abstract

Vision-guided pick cells fail silently when the believed camera pose is wrong: the pose estimator stays confident while every grasp lands off target. In a simulated workcell with two embodiments, a UR10e and a FANUC M-710iC/50 loaded from FANUC's official description, we study a reliability loop around a classical depth localiser. It combines a fixture-residual drift monitor, robust SE(3) recalibration accepted only if it predicts a held-out reference sphere, a learned pose-risk gate against measured grasp tolerances, and collision-aware grasp selection. Every comparison is pre-registered, paired and run on held-out seeds. A 0.5° camera bump drops sorting success from 50/50 to 7/50. The monitored system recovers 42/50 ($p = 6 \times 10^{-11}$) and cuts wrong picks from 251 to 0; its remaining failures are explicit deferrals. Slow drift is detected at a median of 0.1° with zero false alarms in 1,000 frames. Flat fixtures leave camera translation weakly observable (a 10 mm bump is still 7.5 mm off), and the information-matrix condition number does not flag bad fits; reference spheres and held-out validation fix both. In touching-part clutter, fine-tuned Mask R-CNN plus clearance-ordered grasping sorts 88/100 trays, equal to oracle poses with the same rule. The same code transfers to the FANUC arm without robot-specific changes. A pose-risk gate helps only when the perception behind it is good: with classical perception it over-defers, with learned segmentation it halves wrong picks and raises success from 21/50 to 31/50. Meta's JEPA world model needs predictor fine-tuning to beat a copy baseline on our cell, and neither it nor our own latent model foresees grasp outcome (probe AUROC 0.72 on the predicted frame vs 0.99 on the real one), so world models are given no authority over actions. Simulation only.

1 Introduction

Industrial pick-and-place cells are calibrated once and then trusted. A bumped camera, a loosened mount or thermal drift silently corrupts every downstream estimate, and a pose estimator's confidence does not see it, because the geometry it fits is internally consistent but placed in the wrong frame. We ask a narrow, measurable question: *can a perception stack recognise when its geometry is untrustworthy, and act on that, without giving up throughput when nothing is wrong?*

We contribute, in simulation: (i) a validated-recalibration protocol whose acceptance test is leave-one-sphere-out prediction, with a drift dose-response on two embodiments; (ii) measurements locating *where* reliability gains come from: wrong actions become deferrals, phantom detections dominate drift failures, and gripper collisions dominate clutter failures; (iii) a pose-risk gate with measured task tolerances, including its failure modes (coverage loss under shift, and over-deferral when perception is weak); (iv) a negative result on world-model action verification, with a probe that isolates the cause. We do not claim novelty for fixture calibration, conformal prediction or JEPA world models individually, nor any performance on hardware.

2 Related work

Camera calibration and drift. Hand-eye calibration [9] estimates the camera-to-robot transform from known motions. We instead monitor an eye-to-hand camera against the workcell's own fixtures and recalibrate online, and change detection in the spirit of CUSUM [6] motivates a commissioning baseline with a fixed alarm rule. **Pose estimation and uncertainty.** Learned 6D estimators such as FoundationPose [10] and distributional methods such as SE(3)-PoseFlow [3] produce poses or pose distributions. Our earlier BELIEF-3D study [4] showed that view-dependent calibration bias, not pose noise, limits grasping. Here we gate actions on predicted task-frame error, calibrated with conformal methods [1]. **Segmentation.** We fine-tune Mask R-CNN [2] on synthetic, domain-randomised images. **World models.** DINO-WM [11] and JEPA world models [7] plan in the latent space of a frozen self-supervised encoder such as DINOv2 [5]. We test whether such a model, reproduced and adapted to our cell, can verify grasp candidates. Simulation uses MuJoCo [8].

3 Setup

Workcell. A table, a pick tray, two bins, three reference spheres on posts, and a fixed RGB-D camera with OAK-D-Pro-like geometry (80° HFOV). Depth noise follows stereo triangulation, $\sigma_z = z^2\sigma_d/(fb)$, with $b = 75\,\text{mm}$, $f \approx 763\,\text{px}$ and an *assumed* $\sigma_d = 0.1\,\text{px}$ (0.3 for "degraded"). **Robots.** UR10e (MuJoCo Menagerie) and FANUC M-710iC/50 (FANUC's kinematics, joint limits and link inertials; our servo gains), sharing one parallel-jaw gripper. With oracle poses both place 300/300 parts (Wilson 95% [0.987, 1]). **Parts and task.** Block, pin and L-bracket CAD primitives; sort 3 (or 6, touching) parts by class. **Protocol.** Dev seeds < 1000 are used for design; every test cell uses registered, disjoint seed ranges. Arms are paired by seed, with exact McNemar tests on episode success.

4 Methods

B1 classical localiser. Depth → world points (believed extrinsics) → tray and height gates → connected components → height-profile class → Huber point-to-SDF planar fit. **Drift monitor.** The residual r is the median |SDF| of static-scene points to the fixture CAD under the believed extrinsics. Alarm at $r > \mu_0 + 6\sigma_0$ from 20 drift-free commissioning frames. **Recalibration.** Huber least squares over a camera-frame $\mathfrak{se}(3)$ increment. A fit is accepted only if, refit without sphere k, it predicts sphere k within 2 mm for at least two spheres; otherwise the cell defers. **Pose risk.** Gradient boosting maps observable features (registration residual, pixel count, valid-depth fraction, calibration residual, class) to grasp-point error. Isotonic regression maps that to $P(e > \tau_{\text{cls}})$, and split / Mondrian conformal bounds are added. Tolerances come from an oracle-offset sweep: $\tau = 15$ (block), 12 (bracket) and 10 mm (pin). **Learned segmentation (B2).** Mask R-CNN (ResNet-50 FPN, COCO weights) fine-tuned on 3,000 synthetic images; the B1 pose fit is applied inside each mask. **Collision-aware selection (CA).** Pad volumes at pre-shape width are scored by their SDF clearance to the *estimated* neighbours; the clearest part is picked first, and pin yaw is searched. **Recovery.** A budgeted state machine (observe, check calibration, recalibrate, localise, evaluate risk, execute, verify, defer), with reason codes and no unbounded loops. **World models.** JEPA-WM (MetaWorld checkpoint, frozen DINOv2 ViT-S/14) with an explicit action and proprio adapter (1 cm units, five sub-steps, clipped before normalisation). WM-A is frozen; WM-B fine-tunes the 17.6 M-parameter predictor. A fallback latent model is an MLP on a global DINOv2 embedding with an outcome head. The verifier ranks a 5×5 grid of grasp candidates by predicted success.

Table 1: Camera drift, UR10e, 50 paired episodes per cell (EXP-M5-001). Wrong picks = executed and failed.

Bump	B1	B3cal	B3cal wrong / deferred
none	50/50	50/50	0 / 0
0.5°	7/50	**42/50**	0 / 8
1°	2/50	30/50	0 / 20
10 mm	6/50	30/50	0 / 20
3°+20 mm	0/50	7/50	0 / 43

5 Results

Noise alone does not separate methods. B1 equals the oracle (100/100 episodes) under nominal and degraded noise on both robots, so we evaluate under injected faults.

Drift (Table 1). The monitored system beats B1 in every drifted cell ($p \leq 0.016$). B1 makes 116–295 wrong picks per cell; B3cal makes 3 in total. As drift grows, failures become deferrals. Detection is not the bottleneck: alarms fire in 50/50 drifted episodes and 0/50 clean ones. Slow drift (a 0→1° ramp over 20 observations, EXP-M5-004) is detected in 50/50 runs at a median of 0.10° (max 0.35°), before B1's grasp-point error exceeds 5.2 mm, with 0 false alarms in 1,000 drift-free frames.

Observability. With only the table and 24–40 mm walls, a 10 mm camera translation is corrected to 7.5 mm; with three reference spheres, to 1.2 mm. The condition number of $J^\top J$ (400–800) did not separate good from bad fits. Held-out sphere validation refuses the 3°/20 mm case instead of applying a 1.4 mm-accurate but unverified correction.

Where drift failures come from. A tilted camera lifts table patches above the height gate, producing phantom "parts". Measured tolerances (10–15 mm) exceed the pose shift of a 0.5° bump, so phantoms, not small pose errors, dominate. The risk model ranks out-of-tolerance estimates with AUROC 0.995 vs 0.754 for the estimator's score, and 0.966 vs 0.767 on real-part detections only (post hoc). Split-conformal 90% bounds reach 0.89 coverage in distribution and **0.71 under shift**.

Full loop. Adding the risk gate to B3cal changes nothing under drift (identical in all four cells, 0 wrong picks). After recalibration there is nothing left for it to stop.

Learned segmentation. COCO weights alone reach 0.001 segmentation mAP. Head-only, partial and full fine-tuning reach 0.854, 0.853 and 0.861 (one seed each; no difference established), and head-only uses 28% of the memory. In clutter (Table 2) B2 raises first-frame localisation from 60.5% to 99.2% and sorting from 30 to 53/100 ($p = 1.5 \times 10^{-5}$), near the 58/100 oracle ceiling, where gripper collisions take over. Clearance ordering lifts B2 from 50 to 88/100 ($p = 7.5 \times 10^{-11}$), equal to the oracle with the same rule. All 42 remaining failed picks are blocks or brackets (a single closing axis) whose

Table 2: Touching-part clutter, 6 parts, 100 paired trays (EXP-M4-002/003; fresh seeds per study).

Arm	Sorted trays	Wrong picks
B1 classical	30/100	355
B2 Mask R-CNN	53/100	284
B0 oracle poses	58/100	269
B2 (fresh seeds)	50/100	232
B2 + CA	**88/100**	55
B0 + CA	88/100	48

Table 3: World-model prediction on held-out UR10e trajectories: mean latent L2 as a multiple of a copy-z_0 baseline.

h	JEPA zero-shot	JEPA WM-B	Fallback
1	2.76	1.08	0.63
2	2.72	0.82	0.52
6	2.68	0.67	0.43

descent is blocked; pins, whose grasp yaw is free, never fail.

The gate needs good perception. In clutter with B1, the gate cuts wrong picks by 64% but defers 32/50 episodes, and success falls from 14 to 5/50 ($p = 0.004$). Recalibrating it on clutter data made it more selective (wrong picks −84%) and deferred more (4/50 success). With B1 most clutter estimates really are bad, and re-observing a static scene adds no information. With B2 (EXP-M9-005) the gate cuts wrong picks from 95 to 48 and raises success from 21 to 31/50 ($p = 0.031$), with 2 deferrals.

Transfer. The same code on the FANUC M-710iC/50: a 0.5° bump gives 6/50 (B1) vs 42/50 (B3cal), $p = 3 \times 10^{-11}$; 10 mm gives 9 vs 30/50. The camera and fixtures are shared, so this measures modular transfer, not generalisation to a new cell.

World models. We reproduced JEPA-WM on Meta-World reach-wall: 5/12 vs the paper's 41.6% [7], with $n = 12$ and a CI of roughly 15–72%. Zero-shot on our cell it is worse than copying the current frame at every horizon (Table 3). Fine-tuning only the predictor brings it to 0.67× at $h = 6$. The fallback predicts a global embedding, an easier target, and reaches 0.43×. As a grasp verifier the fallback *hurt*: 88/100 vs 99/100 for geometry at zero drift ($p = 0.001$), and 16 vs 21/100 at 1°. Its outcome head reached only AUROC 0.58–0.67, below a pre-set 0.80 gate. A probe on JEPA patch latents isolates the cause: grasp outcome is readable from the *true* final frame (AUROC 0.99) but not from WM-B's *predicted* one (0.72). The models learn the arm's motion, not the contact outcome.

Deployment. The fine-tuned Mask R-CNN exports to ONNX with exact outputs (76/76 detections, $|\Delta\text{score}| \leq 1.8 \times 10^{-6}$), but ONNX Runtime is about 12× slower than PyTorch on a V100 (544 vs 44 ms), and its TensorRT provider slower still, so PyTorch is kept. A ROS 2 Jazzy integration test (sim bridge, perception, calibration monitor) passes.

6 Limitations

Everything is simulation, with assumed sensor noise and primitive parts. Drift is a rigid bump or a linear ramp of the camera, and real mounts also flex. Training runs use single seeds. The vision domain shift we built was too mild to probe generalisation. World-model data comes from a scripted policy. JEPA-WMs is licensed CC BY-NC. No result here establishes performance on real FANUC, UR or any vendor's software.

7 Conclusion

Reliability came from three things, each measured separately: detecting when geometry is wrong (validated recalibration), refusing actions that are likely wrong (risk gating, *if* perception is good enough), and choosing actions that are less likely to fail (clearance-ordered grasping). Learned components earned their place where they fixed a specific failure (merged blobs) and were kept away from decisions they could not yet make (world-model verification). The experiment registry, code and all result files are public at `https://github.com/parammadan/aura3d`.

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